Try it and explain your thinking

Explain and apply the idea

Create a five-column authority map: owner, designer, contractor QC, owner QA/acceptance and regulator. Place MMCD/ministry/project documents in their actual roles.

Skills you use: Reading, Writing, Problem solving.

Calculate and check the result

Write a clarification request for the worked case that names the conflict, location, current versions and decision needed.

Skills you use: Writing, Communication.

Communicate and verify the work

Build a digital revision register and peer-test whether another learner can find the current entrance requirements. Add a new response and show the superseded record without deleting its history.

Skills you use: Digital, Collaboration, Adaptability, Creativity and innovation.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Use a permitted project document set or the complete fictional case.	0 / 1 / 2	Evidence/help: _____
Assess correct document selection, interpretation and referral of an unresolved design conflict.	0 / 1 / 2	Evidence/help: _____
No classroom interpretation changes a real contract or design.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, C4, printed p. 40; source PDF pp. 42–42.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 2: Petroleum Asphalt; source PDF pp. 55–142. Open publication.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 15: Special Uses of Asphalt and HMA; source PDF pp. 737–802. Open publication.
- BC Ministry of Transportation and Transit. (2025). 2025 Standard Specifications for Highway Construction, Volume 1. Open publication.
- Master Municipal Construction Documents Association. (Web snapshot: October 30, 2026). Supplementary Documents. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Study chapter · Outline D1

Environmentally friendly properties

An environmental claim is useful when it names the material, exposure conditions, service provided and evidence. Asphalt can be recovered and used again, but 'recyclable' is not the same as 'recycled here', and neither word proves the lowest overall impact. The learning goal is a defensible comparison that respects the people, land and water affected by a project.

Define Green properties of asphalt

LEARNING STEP 1

Define Green properties of asphalt

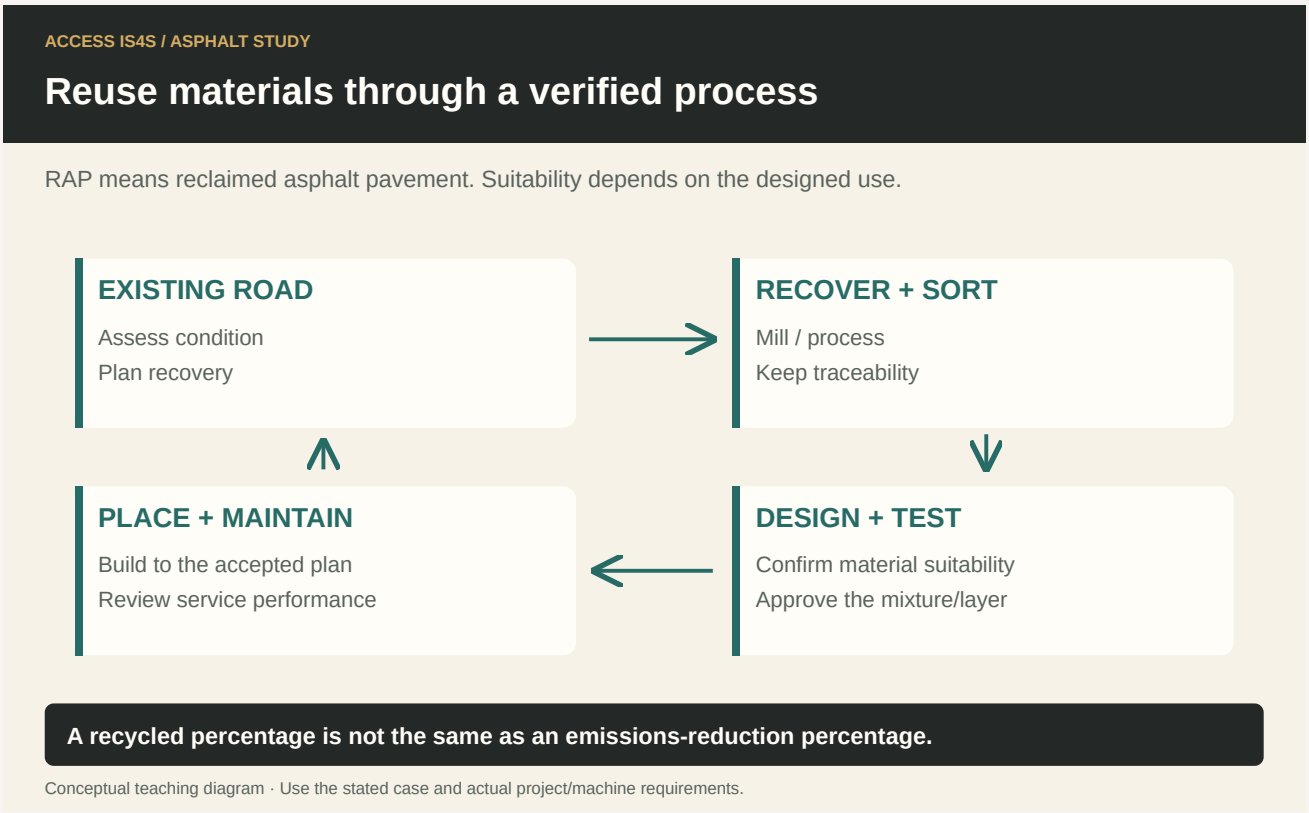
Reclaimed asphalt pavement (RAP) contains recovered aggregate and aged binder. Processing and an approved mix or other permitted use can recover value from those materials. The receiving process must account for variability, contamination, binder properties and the required final performance. A recycling pathway has collection, processing, transport and production effects of its own.

Avoid blanket claims that all installed asphalt is 'non-toxic' or 'does not leach' under every condition. Product composition, contamination, exposure, receiving environment and applicable criteria determine what evidence is needed. An intact pavement, hot production process, milling dust and a stored pile are different exposure situations. A claim about one does not automatically apply to the others.

A fair comparison defines the same required service, area/loading, service period and maintenance assumptions. Include production, transport, construction, use/maintenance and end-of-life within the stated boundary. A lower production impact per tonne can be outweighed by different quantities, travel or service life. Environmental decisions can also involve water, noise, land, access and local priorities that a single emissions number does not capture.

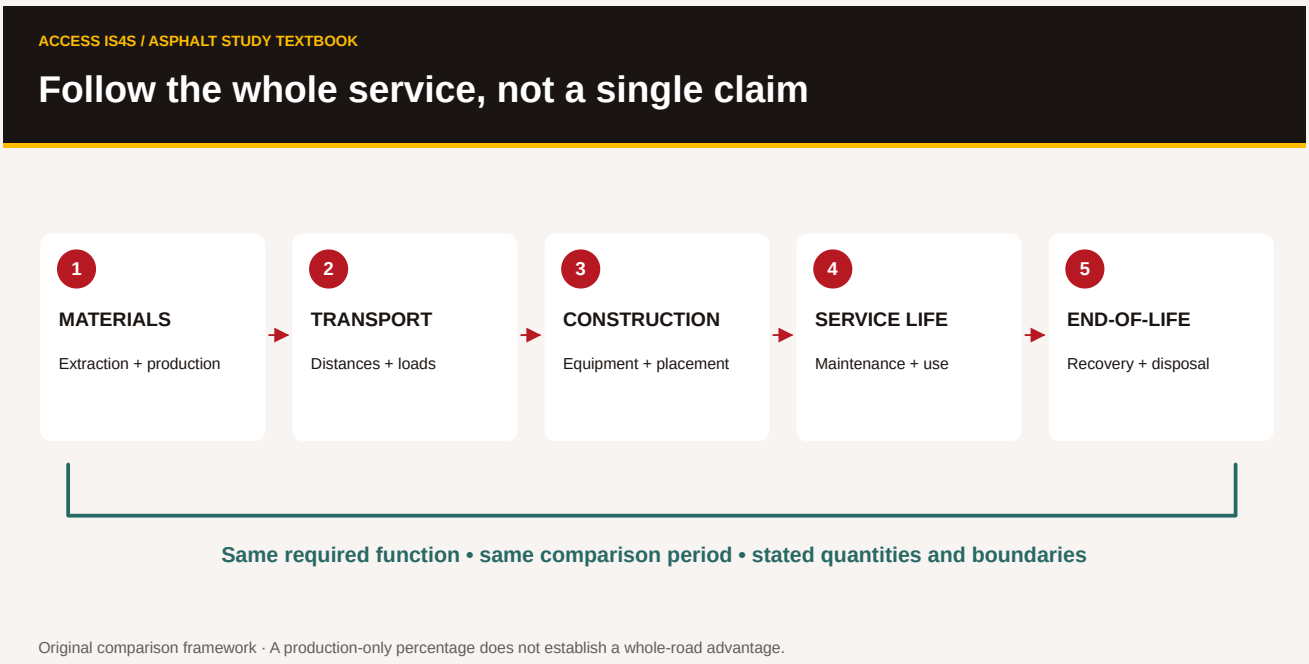
In B.C., meaningful local engagement identifies the appropriate people and decision processes; a class cannot supply a Nation’s permission or knowledge. Use public/fictional evidence for the exercise and invite contributions only by choice. A careful conclusion can acknowledge a documented reuse pathway while asking for the data needed to compare actual options.

Reuse materials through a verified process



Note. ACCESS IS4S. (2026). Reuse materials through a verified process [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

Follow the whole service, not a single claim



Note. ACCESS IS4S. (2026). Follow the whole service, not a single claim [Original instructional SVG]. Conceptual illustration; read the stated conditions in the lesson.

Worked example · Recycled proportion is not a full impact score

Fictional material record: a 240 t proposed mix contains 48 t of approved RAP. A sales note calls it “20% recycled, therefore 20% lower whole-life emissions.” No emissions or service-life data is supplied.

Think it through.

48/240 × 100 = 20% by mass in this record. That establishes the stated proportion only. Whole-life emissions need comparable quantities, processing/production, transport, construction, maintenance and end-of-life evidence. The proposed reduction is not demonstrated by the recycled percentage.

Try it and explain your thinking

Explain and apply the idea

Rewrite the sales note into one supported statement and three questions needed for a whole-life comparison.

Skills you use: Reading, Writing, Problem solving.

Calculate the RAP mass for a second fictional 300 t mix at 15%

Calculate the RAP mass for a second fictional 300 t mix at 15%. Explain why comparing 48 t with the second mass does not rank environmental performance.

Skills you use: Numeracy, Communication.

Create a visual comparison brief with explicit boundaries

Create a visual comparison brief with explicit boundaries. Test it with a partner who asks about runoff or community access, then revise and save it.

Skills you use: Digital, Collaboration, Adaptability, Creativity and innovation.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Assess evidence interpretation using documented or fictional product records.	0 / 1 / 2	Evidence/help: _____
Check that the learner distinguishes recovery potential, actual reuse and verified environmental outcomes.	0 / 1 / 2	Evidence/help: _____
Environmental approvals and material-use decisions remain with the appropriate qualified/authorized processes.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, D1, printed p. 41; source PDF pp. 43–43.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 13: Rehabilitation Methods for Asphalt Pavements; source PDF pp. 653–710. Open publication.
- Province of British Columbia. (Web snapshot: October 30, 2026). Asphalt Plant Regulation, B.C. Reg. 217/97. Open publication.
- Province of British Columbia. (Web snapshot: October 30, 2026). Report a Spill. Open publication.

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Study chapter · Outline D2

Use of storage areas

Storage keeps the right material identifiable and available while controlling release, incompatibility and exposure. A storage plan includes the container/tank, surrounding area, access, inspection, emergency response and destination for waste. A tidy-looking tank is not proof that its containment or product identity is adequate.

- Type 1 storage
- Type 2 storage
- Use of recycled materials in hot mix asphalt
- Use of recycled materials in asphalt plant burners

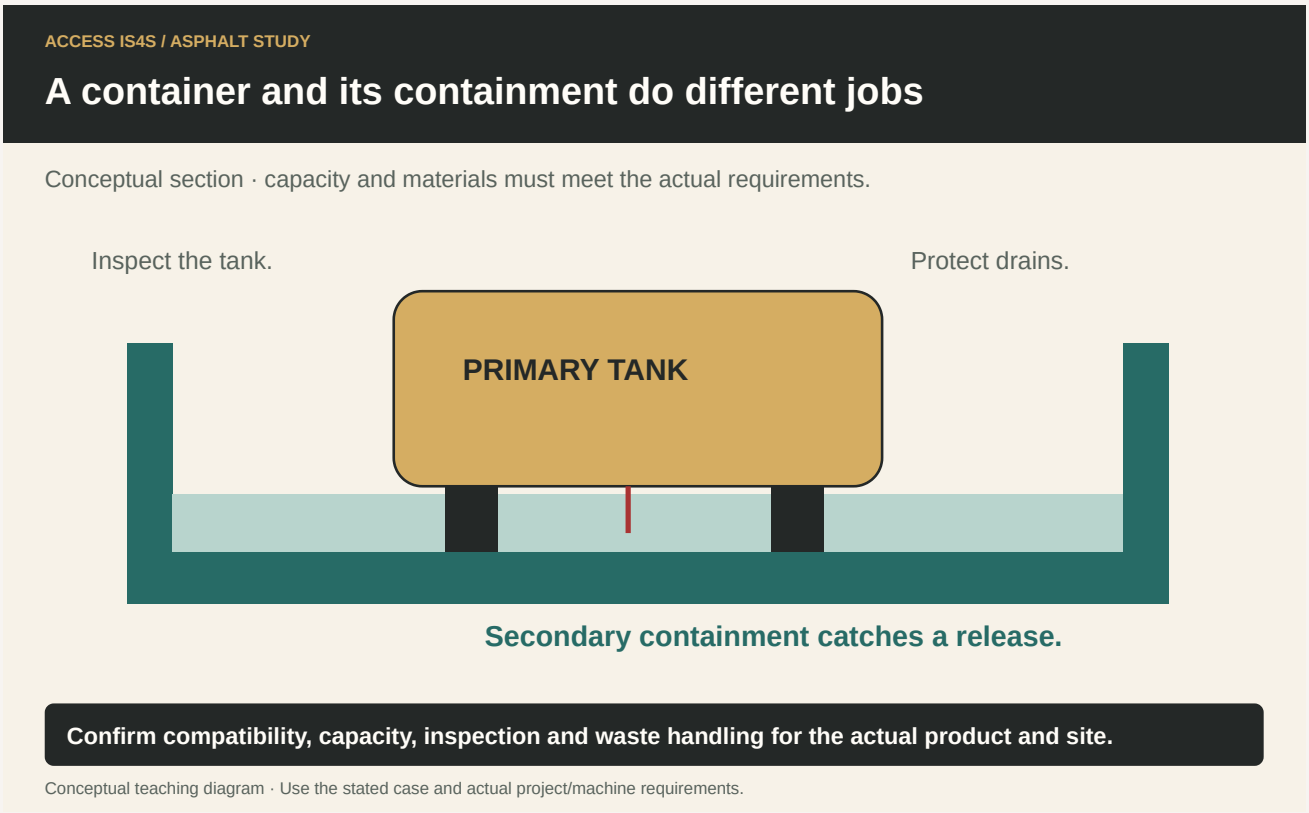
LEARNING STEP 1

Type 1 storage

For this lesson, **Type 1** describes a primary storage container with a separate spill-containment arrangement, such as a tank within a designed containment area. It is a course grouping, not a universal legal classification. Fuels, primers, emulsions such as an identified SS-1 product, asphalt binder and waste streams require their actual SDS, compatibility, heating/handling and regulatory arrangements.

Identify the product and hazards before deciding which area is suitable. Natural gas service has its own engineered supply and safety requirements; it is not handled as an ordinary open liquid tank. Containment must remain effective, including when rainwater or other material occupies space. Drainage and disposal of collected liquid follow the approved procedure; opening a drain to clear capacity can release contamination.

A container and its containment do different jobs



Note. ACCESS IS4S. (2026). A container and its containment do different jobs [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

LEARNING STEP 2

Type 2 storage

Type 2 here describes a self-contained storage arrangement, such as equipment with integral secondary containment, where its design provides that function. The word self-contained does not mean leak-proof, maintenance-free or automatically suitable for every fuel or liquid asphalt product. Verify the actual design, product compatibility, capacity, inspection and permitted location.

A biodiesel blend can have different handling/storage considerations from another fuel. A heated binder storage system introduces thermal and other hazards. Labels, isolation, overfill prevention, transfer connections and emergency access must match the actual installation. Read the tank/assembly instructions and product records together.

LEARNING STEP 3

Use of recycled materials in hot mix asphalt

Using RAP in hot mix can reduce demand for some new material and may offer cost benefits, but the outcome depends on processing, transport, consistency, binder contribution and required performance. Mobile recycling changes where processing occurs and may change traffic, emissions and material-control

arrangements. Identify what is being recycled and into which approved product; do not treat every recovered pile as acceptable feed.

Keep recovered material segregated by the required source/type and protect it from unacceptable contamination. The quality record follows it through processing and blending. A low purchase price alone is not the cost of a compliant finished mixture.

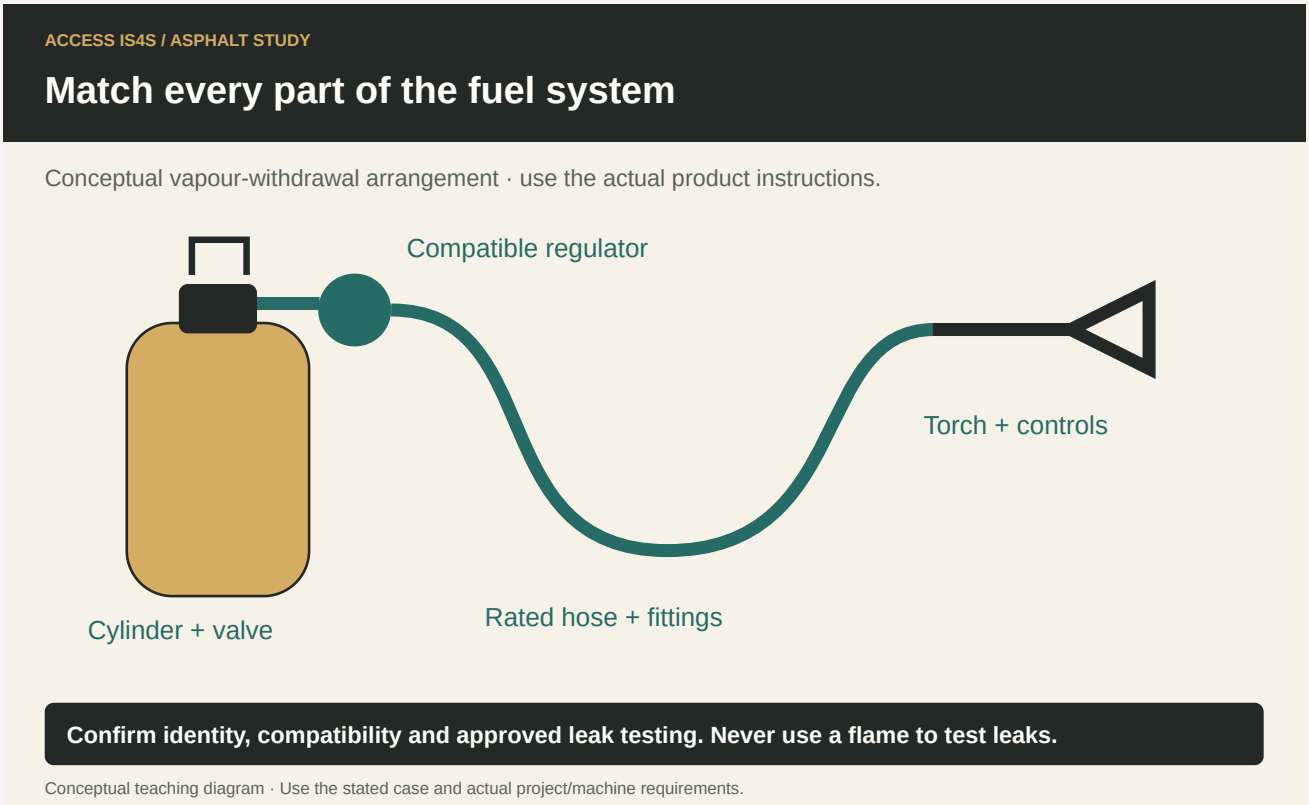
LEARNING STEP 4

Use of recycled materials in asphalt plant burners

A plant burner must use only fuel permitted by its design and applicable environmental authorization. Recovered fuel or other recycled material is not automatically acceptable because it burns. Composition, contamination, handling, burner suitability, combustion control and emissions requirements matter. Heating temperature is established by the approved process and equipment, not by a generic classroom number.

Storage/transfer for a permitted recovered fuel needs the actual SDS and emergency arrangements. A change in fuel can affect burner performance and emissions and must be controlled through the authorized process. Do not conduct a classroom trial with an unidentified waste fuel.

Match every part of the fuel system



Note. ACCESS IS4S. (2026). Match every part of the fuel system [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

Worked example · Containment is a working system

Fictional inspection record: the containment area is intact, but its drain is open and collected rainwater has not been assessed for contamination. A delivery is due in 30 minutes.

Think it through.

The record does not establish effective containment for the transfer. Have the responsible person implement the storage/transfer procedure, address the open drain and assess the collected liquid before release or transfer. Delivery timing does not change the product or environmental requirements. Record the condition and resolution rather than ticking “tank intact” as the whole inspection.

Try it and explain your thinking

Draw the difference between separate containment and an integral self-contained arrangement

Draw the difference between separate containment and an integral self-contained arrangement. Identify five checks both still require.

Skills you use: Reading, Communication.

Calculate and check the result

Fictional cost exercise: virgin material supply costs $8,000; recovered feed costs $5,000 plus $2,400 processing and $900 extra transport. Compare only these stated costs.

Skills you use: Numeracy, Writing, Problem solving.

Create a digital storage inspection record and peer-test it with the open-drain case

Create a digital storage inspection record and peer-test it with the open-drain case. Revise it to distinguish observation, action, responsible person and verified closure.

Skills you use: Digital, Collaboration, Adaptability, Creativity and innovation.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Use the actual installation, SDS, transfer plan and permitted observation area.	0 / 1 / 2	Evidence/help: _____
Assess product identification, containment/compatibility questions and reporting of a defect.	0 / 1 / 2	Evidence/help: _____
Live fuel/binder transfer and burner changes require separate authorized instruction and controls.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, D2, printed p. 42; source PDF pp. 44–44.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 7: Manufacture of Hot Mix Asphalt; source PDF pp. 341–390. Open publication.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 5: Chemical and Biological Substances. Open publication.
- Province of British Columbia. (Web snapshot: October 30, 2026). Report a Spill. Open publication.
- Canadian Centre for Occupational Health and Safety. (retrieved October 30, 2026). Chemical Profiles: Propane. Open publication.

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Study chapter · Outline D3

Asphalt plant and placement emissions

Emissions are releases from a process into the environment; worker exposure is contact with a hazard at the workplace. They are related but are not measured or regulated in exactly the same way. A stack result does not establish a worker's breathing-zone exposure, and a workplace mask does not establish compliance with an environmental permit.

- Discuss role of Metro Vancouver (formerly GVRD) in enforcement and regulation of emissions
- Role of MOE in establishing regulations
- Use of Bag house
- Emission controls

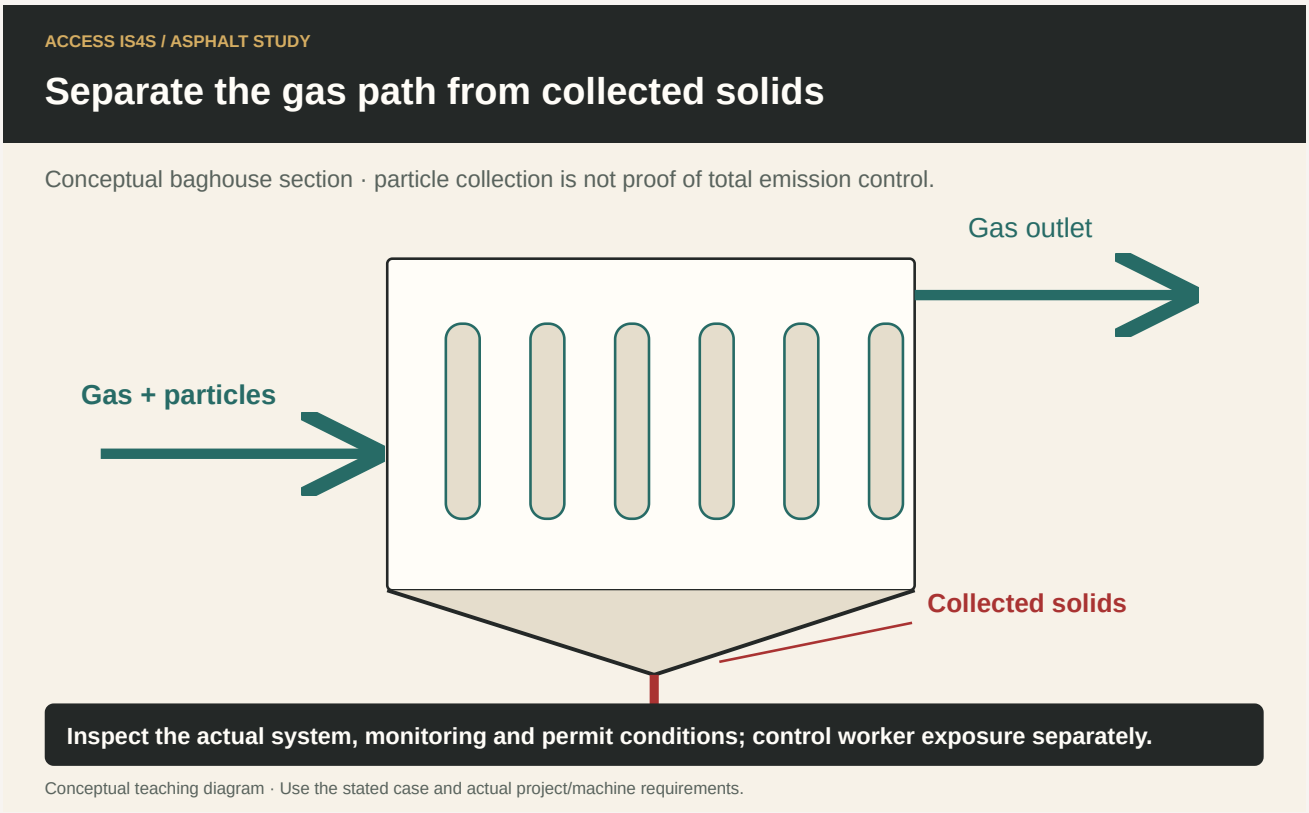
LEARNING STEP 1

Discuss role of Metro Vancouver (formerly GVRD) in enforcement and regulation of emissions

Metro Vancouver has a regional air-quality regulatory program. Determine whether the plant or activity is within that jurisdiction and which permit, bylaw or authorization applies. The retained B.C. Asphalt Plant Regulation explicitly states that it does not apply in the Metro Vancouver Regional District. Do not carry an outside-region authorization into the region by assumption.

The practical record identifies the actual site, operator, authorization, operating conditions, monitoring/reporting requirements and responsible contacts. A historical count of member communities is not a compliance test. Consult the current regional program and actual permit when preparing the work.

Separate the gas path from collected solids



Note. ACCESS IS4S. (2026). Separate the gas path from collected solids [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

LEARNING STEP 2

Role of MOE in establishing regulations

Outside Metro Vancouver, the B.C. Asphalt Plant Regulation establishes requirements for the covered hot-mix and hot-in-place recycling plants, with applicable registration, emission, operating and reporting provisions. Its text also addresses additional municipal/regional requirements. Identify the actual activity and location before selecting the rule. Provincial ministry names and responsibilities can change; the legal instrument and current authorized contact must be recorded.

An environmental authorization is different from the workplace exposure-control program. Both can be relevant to the same plant. Keep monitoring data and reports associated with their method, location, interval and required interpretation.

LEARNING STEP 3

Use of Bag house

A **baghouse** is a fabric-filter dust collection system. In the MS-4 concept, dust-laden gas passes through the filtering arrangement, particles collect and cleaning releases collected material to a hopper. The gas and collected-solids paths are different. Primary collection may remove larger particles before secondary collection. The actual plant arrangement determines how fines are returned or removed.

Filter condition, seals, airflow and the designed cleaning/collection system affect performance. A visible change, alarm or damaged component needs the operator's prescribed response. Do not assume a baghouse removes every gaseous pollutant: its principal dust-control function does not establish control of all emissions.

LEARNING STEP 4

Emission controls

Burner operation, aggregate moisture, production rate, material temperature, handling and dust-control performance can influence emissions. Heating asphalt excessively or using an unsuitable fuel can create problems; actual limits and operating settings come from the approved plant process and authorization. Placement work also needs exposure controls suited to the product and conditions.

Record observations without diagnosing the entire process from colour or odour alone. An unusual visible plume can prompt the required response and investigation, but it is not a measured concentration. Keep the operating interval and material/fuel identity so the responsible people can relate the observation to production and monitoring records.

Internal components of a baghouse



Note. Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4, Fig. 7.13, p. 324; PDF p. 358). Faithful scan crop; original figure caption retained. Original image contributor: Terex Roadbuilding; credit register, PDF pp. 803–805. Full source context.

Worked example · Two measurements, different questions

Fictional records show a compliant stack sample from last week and a worker complaint about fumes at today's loadout. The supervisor proposes closing the complaint using the stack result.

Think it through.

The records differ in location, time and purpose. The stack result does not resolve today's worker exposure. Apply the workplace reporting/assessment process, control the affected task as required and retain both records with their identities. Investigate the actual loadout conditions and any process change without inventing a concentration.

Try it and explain your thinking

Trace dirty gas, filtered gas and collected dust through the baghouse drawing

Trace dirty gas, filtered gas and collected dust through the baghouse drawing. Explain one pollutant-control claim the drawing cannot support.

Skills you use: Reading, Communication.

Create a jurisdiction-and-record table for a fictional plant inside Metro Vancouver and one outside

Create a jurisdiction-and-record table for a fictional plant inside Metro Vancouver and one outside. Keep environmental and worker-exposure records in separate columns.

Skills you use: Writing, Problem solving.

Save a plume/complaint observation form

Save a plume/complaint observation form. A partner adds a change of fuel and a different time interval; revise the questions and explain the missing evidence.

Skills you use: Digital, Collaboration, Adaptability, Creativity and innovation.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Use actual site authorization and exposure programs under qualified supervision.	0 / 1 / 2	Evidence/help: _____
Assess correct record/jurisdiction selection and explanation of dust-control components.	0 / 1 / 2	Evidence/help: _____
Emission testing, permit interpretation and occupational exposure assessment require the applicable competent professionals.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, D3, printed p. 43; source PDF pp. 45–45.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 7: Manufacture of Hot Mix Asphalt; source PDF pp. 341–390. Open publication.
- Province of British Columbia. (Web snapshot: October 30, 2026). Asphalt Plant Regulation, B.C. Reg. 217/97. Open publication.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 5: Chemical and Biological Substances. Open publication.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 6: Substance Specific Requirements. Open publication.

- Metro Vancouver. (n.d.). Air Quality Regulatory Program. Current program locator, accessed 30 October 2026; verify actual permit requirements. Open publication.

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Study chapter · Outline E1

Determine asphalt mix design

A **mix design** establishes a suitable combination of aggregate, binder and other permitted ingredients for the required pavement performance. The accepted **job mix formula (JMF)** then provides the production target under the contract. It is more than a recipe by eye: the materials, proportions, test basis and performance requirements belong together.

- Asphalt/aggregate composition
- Types of mix designs
- Mix temperature characteristics

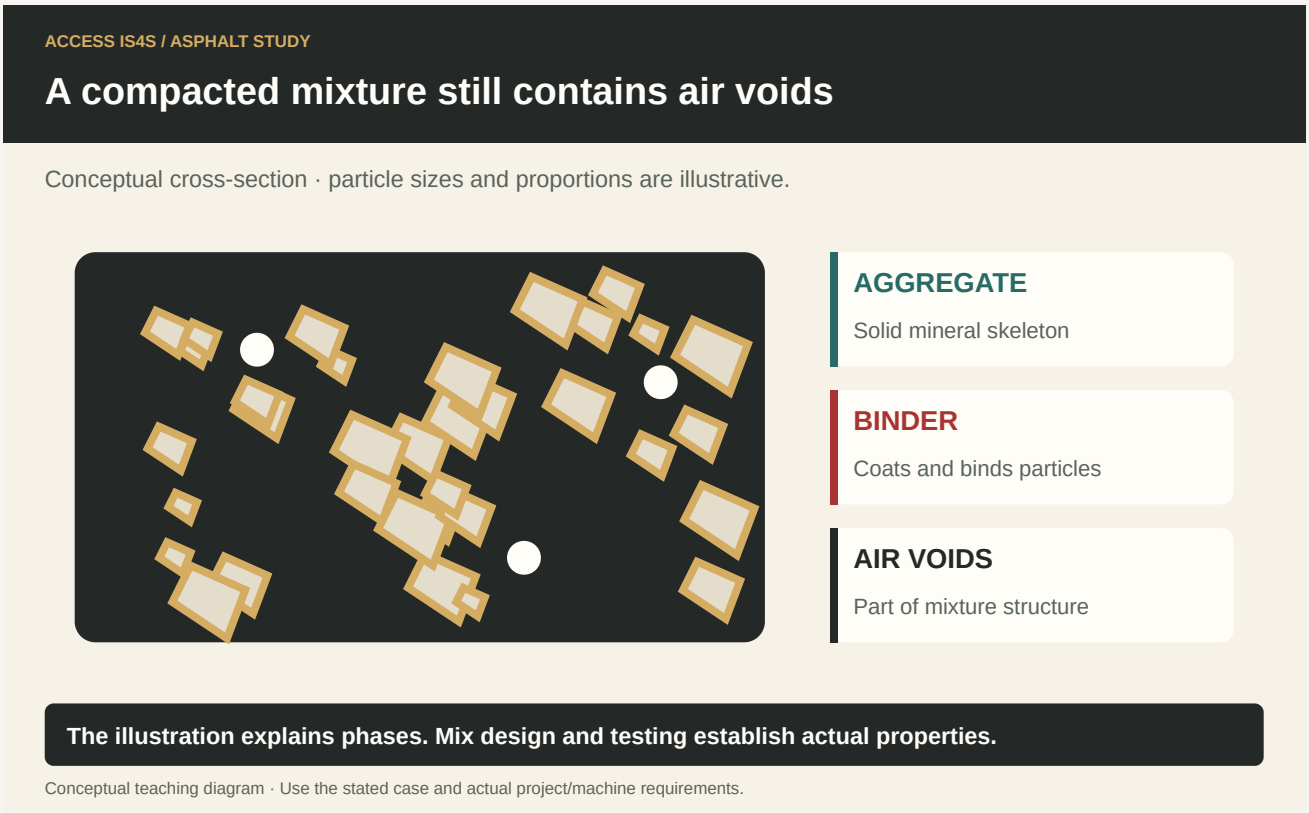
LEARNING STEP 1

Asphalt/aggregate composition

The aggregate skeleton, binder and air voids form a mixture whose response depends on both materials and compaction. **Marshall** design uses its specified specimen preparation and stability/flow and volumetric evaluations. **Superpave** design uses its specified material selection, gyratory compaction and volumetric framework, with additional tests where required. Their results and specimen methods are not interchangeable labels.

A field crew does not redesign a mixture by adding binder when the surface looks dry. Document the observation, identify the supplied JMF and refer the condition through quality control. Changes in aggregate source, binder grade or recovered material can affect the design and may require the specified approval process.

A compacted mixture still contains air voids



Note. ACCESS IS4S. (2026). A compacted mixture still contains air voids [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

LEARNING STEP 2

Types of mix designs

Dense-graded, open-graded, gap-graded and other designed mixtures distribute aggregate sizes differently and may serve different functions. A mixture chosen for drainage is not automatically suitable for every structural or traffic demand. The specification identifies the required mixture type, nominal size, binder and other design properties; the accepted formula identifies the actual approved combination.

Read the formula's product/source identities, target proportions, binder grade/content, recovered material where permitted, test basis and version. A production ticket with the wrong JMF identifier is a traceability issue even if the mixture looks familiar. Keep design approval separate from daily production conformity.

LEARNING STEP 3

Mix temperature characteristics

Hot-mix asphalt is produced and placed under its required elevated-temperature process. Warm-mix technologies permit specified production/placement approaches at reduced temperatures relative to the corresponding conventional process. Cold-mix products use a different material/process arrangement and are selected for the intended application. These categories do not supply one universal upper/lower temperature.

The approved mix, binder, equipment, haul, ambient conditions and placement/compaction procedure determine the usable temperature requirements. Read actual values from the governing documents and measure by the specified method. A surface infrared reading and an internal probe reading answer different questions. Temperature is recorded with location, time, method and instrument identity.

Marshall compaction hammer

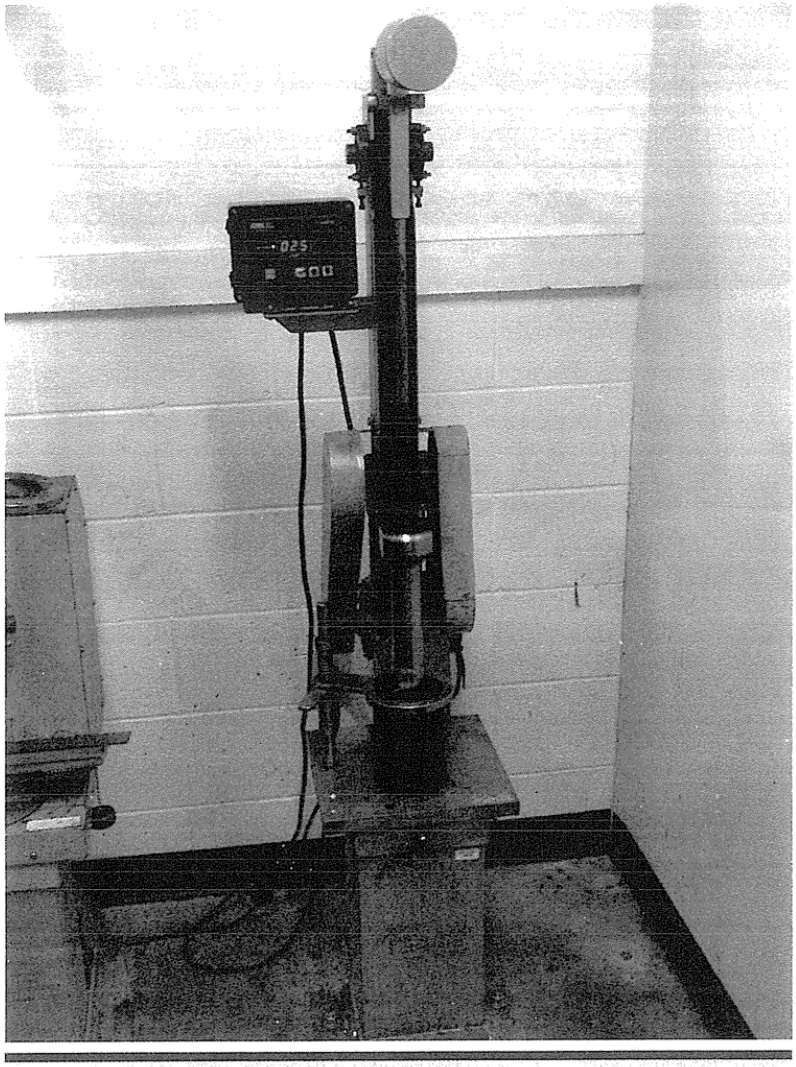


Figure 4.18 Marshall Compaction Hammer

Note. Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4, Fig. 4.18, p. 199; PDF p. 233). Faithful scan crop; original figure caption retained. Full source context.

Worked example · Keep the percentage basis

Fictional teaching JMF uses 5.2% binder by total mixture mass. A 25.0 t batch is considered for arithmetic only. No production temperature or field design is prescribed.

Think it through.

Binder model mass = 25.0 × .052 = 1.30 t; aggregate/other stated remainder = 23.70 t if no other components are separately specified. If the percentage were instead based on aggregate mass, this calculation would be different. Confirm the stated basis before comparing records.

Try it and explain your thinking

Explain Marshall and Superpave as different specified design/testing frameworks

Explain Marshall and Superpave as different specified design/testing frameworks. List five JMF fields needed by a production/placement crew.

Skills you use: Reading, Communication.

Recalculate binder model mass for 32.0 t at 5.2%

Recalculate binder model mass for 32.0 t at 5.2%. Explain why changing the mass percentage in the field is not an authorized design change.

Skills you use: Numeracy, Writing, Problem solving.

Build a digital JMF-reading card and have a partner swap the version identifier

Build a digital JMF-reading card and have a partner swap the version identifier. Revise the delivery check and record what must be confirmed.

Skills you use: Digital, Collaboration, Adaptability, Creativity and innovation.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Use an authorized real JMF or the complete fictional arithmetic card.	0 / 1 / 2	Evidence/help: _____
Assess accurate identification of mixture, design method, percentage basis and temperature measurement requirements.	0 / 1 / 2	Evidence/help: _____
Mix design approval and changes are assessed by the qualified design/quality process.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, E1, printed p. 44; source PDF pp. 46–46.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 4: Hot Mix Asphalt (HMA) Design; source PDF pp. 193–280. Open publication.
- BC Ministry of Transportation and Transit. (2025). 2025 Standard Specifications for Highway Construction, Volume 1. Open publication.
- BC Ministry of Transportation and Transit. (2025). 2025 Standard Specifications for Highway Construction, Volume 2. Open publication.

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Study chapter · Outline E2

Confirm that samples match mix design

A test result belongs to a sample, and a sample must represent an identified material and interval. **Sampling plan**, **increment**, **composite sample**, **sample reduction**, **container**, **chain of custody** and **test method** are connected terms. A precise laboratory number is not useful if the wrong material was sampled or its identity was lost.

- Aggregate sampling
- Belt sampling
- Lab tests
- The process of taking mix temperature readings

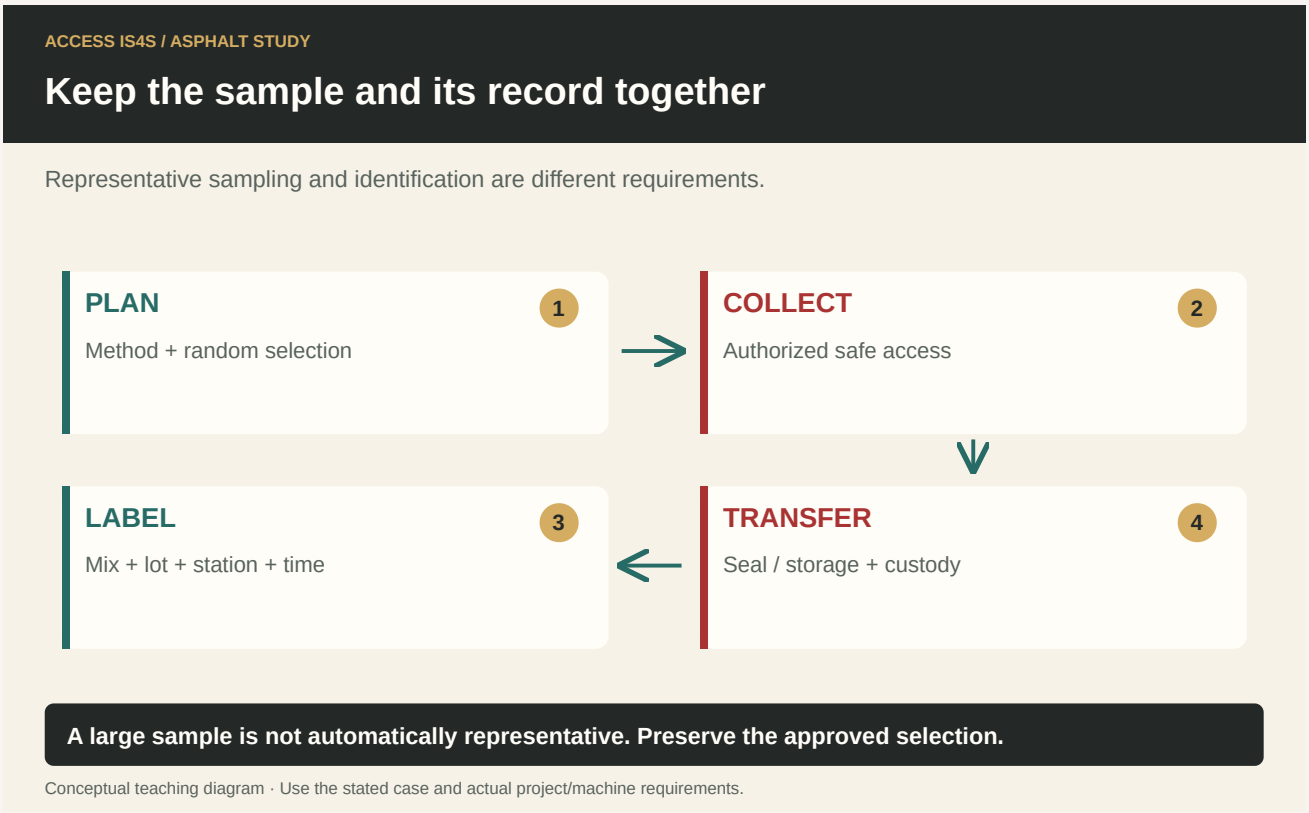
LEARNING STEP 1

Aggregate sampling

Aggregate sampling must follow the specified location, timing, number of increments and method. A convenient scoop from the visible top of a pile may not represent the material being supplied. Weigh-bin or other plant sampling requires the designed safe access and procedure; exposure to hot material, gravity and connected equipment must be controlled.

Preserve product/source identity and the relevant production interval. Combine or reduce samples only as the method permits. Record unusual conditions and contamination. A sample is not made representative by increasing its label detail after an unsuitable collection.

Keep the sample and its record together



Note. ACCESS IS4S. (2026). Keep the sample and its record together [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

LEARNING STEP 2

Belt sampling

Belt sampling can characterize a material stream when performed under the approved method and safe arrangement. A worker must not reach across or into a live moving belt. The required isolation/access and sampling device/procedure determine how the sample is obtained. Include the full required stream/fraction rather than selecting the pieces that look most typical.

A drum-plant process record connects the sampled feed with its bin, proportioning and time. A stopped-belt method and an approved automatic sampling device have different controls; neither is replaced by a generic drawing. Keep the sample method and the safe-work method together.

LEARNING STEP 3

Lab tests

Laboratory tests can determine gradation, binder content and mixture volumetric or other specified properties. The result must identify the method, units, sample and applicable comparison. Method differences and corrections can affect interpretation. A result outside a limit prompts the defined quality process; it does not justify editing the result or replacing it with a more convenient sample.